

For Humans

What the Riemann Hypothesis Means

Diego Rincón
Independent

Abstract

This paper is for people who are not mathematicians. The Riemann Hypothesis is one of the most famous unsolved problems in mathematics. It is about prime numbers. It says the primes are arranged as regularly as possible given that they are arranged at all. It is probably true. No one has proved it. This paper says what it means, why it matters, and why it is hard.

1 What Prime Numbers Are

A prime number is a number that cannot be divided evenly by any smaller number except 1. The first few primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, ...

Every whole number is a product of primes. For example: $12 = 2 \times 2 \times 3$. This decomposition is unique: each number has exactly one prime factorization. The primes are the atoms of arithmetic.

2 How Primes Are Distributed

As numbers get larger, primes get rarer. Near 1000: about 1 in 7 numbers is prime. Near 1,000,000: about 1 in 14. Near 1,000,000,000: about 1 in 21. The pattern: near N , approximately 1 in $\ln N$ numbers is prime. This is the prime number theorem, proved in 1896.

But the primes are not perfectly regular. They cluster and gap in ways that seem irregular. The twin prime conjecture asks: are there infinitely many pairs like $\{11, 13\}$ or $\{17, 19\}$? Goldbach's conjecture asks: is every even number the sum of two primes? These questions remain open.

3 The Riemann Hypothesis

In 1859, Riemann introduced a function $\zeta(s)$ (the “zeta function”) that encodes everything about the primes. It is defined for complex numbers s , and its behavior mirrors the distribution of primes.

The zeros of ζ — the values of s where $\zeta(s) = 0$ — control how evenly the primes are distributed. Each zero introduces a slight irregularity in the count of primes.

Claim 1 (The Riemann Hypothesis). All the “interesting” zeros of ζ lie on the vertical line $\text{Re}(s) = 1/2$ in the complex plane.

This line is called the *critical line*. The hypothesis says: the zeros are perfectly arranged, all on one line, not scattered off it.

4 Why It Matters

If RH is true, the primes are as regular as they could possibly be. The count of primes up to N differs from its expected value by at most about $\sqrt{N} \cdot \log N$ — small relative to N . This is the best possible error estimate.

If RH is false, some zeros are off the critical line. Off-line zeros create larger irregularities in the primes. The primes would be less predictable, less structured.

RH is the statement that the universe of arithmetic has a deep order. Not perfect order — the primes are not perfectly regular — but optimal order: as much order as their definition allows.

5 Why It Is Hard

Note 1 (The wall). To prove RH, one must show that no zero can be off the critical line. This requires ruling out something we have never seen.

The BN approach (the subject of this body of papers) reformulates RH as a question about approximation: can the indicator function $1_{(0,1)}$ be approximated arbitrarily well by combinations of integer-based functions? This is equivalent to RH, but no easier. The wall is always there.

Note 2 (What is known). We know: there are no zeros with $\text{Re}(s) > 1$ (trivially). We know: there are no zeros with $\text{Re}(s) = 1$ (proved 1896; needed for PNT). We know: the first 10^{13} zeros are on the critical line. We suspect: they all are. We cannot prove it.

Note 3 (What would change if RH were proved). Hundreds of theorems in number theory say “assuming RH, ...” These would all become unconditional results. New tools for cryptography (which relies on the unpredictability of primes). A deeper understanding of why the integers are structured the way they are. And: the Clay Mathematics Institute would award \$1,000,000.

6 What This Body Says

This body ([1]) argues that RH is not just a statement about zeros. It is a statement about knowledge: whether the arithmetic of the integers — all the facts about multiplication, divisibility, primes — is sufficient to determine the full analytic structure of ζ .

The body says: the primes teach us about ζ , step by step. Each prime we learn improves our approximation. RH is the claim that this teaching, continued forever, eventually reaches the truth.

The wall is the moment where we have learned everything arithmetic can teach and still do not know whether it is enough. The Riemann Hypothesis says: it is.

References

- [1] Rincón, D. (2026). Wisdom. Preprint.
- [2] Rincón, D. (2026). Numerics. Preprint.
- [3] Rincón, D. (2026). Prime. Preprint.